EVALUATING LIMITS

Dividing by the highest power

This technique isolates the dominant terms that control behavior at infinity.

01

INFINITY OVER INFINITY — BEFORE

$$\lim_{x\to\infty} \frac{3x^2+5x-1}{2x^2-7}$$

↓ Divide every term by x^2 — top and bottom by the same nonzero quantity: the value does not change

02

DIVIDED BY X^2

$$\lim_{x\to\infty} \frac{3+\frac{5}{x}-\frac{1}{x^2}}{2-\frac{7}{x^2}}$$

↓ Let $x \to \infty$ — every term with x in the denominator vanishes

03

LEADING COEFFICIENTS — AFTER

$$\frac{3+0-0}{2-0} = \frac{3}{2}$$

▼ WHY NOT 1?